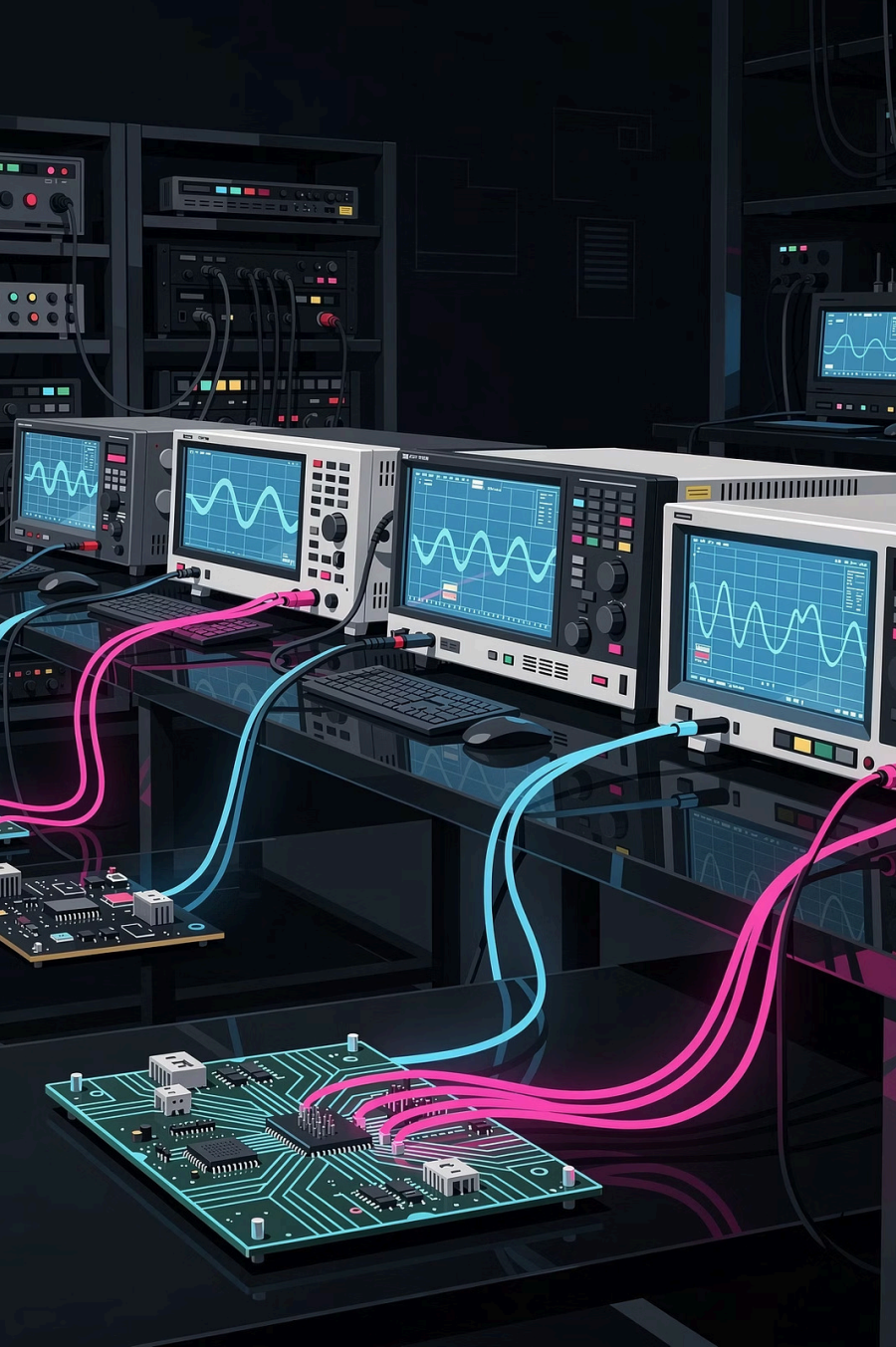




Market Sizing Analysis

White Rabbit Timing System · Beam Position Monitor · Low Level RF

SAFRAN | U.S. MARKET STRATEGY

Agenda

This analysis covers three core product lines – White Rabbit, BPM, and LLRF – across customer personas, market scope, and serviceable addressable market sizing.

01

White Rabbit Timing System

Strengths & weaknesses · HEP and Military/Aerospace personas

02

Beam Position Monitor

Strengths & weaknesses · Scientific Labs and Healthcare personas

03

Low Level RF

Strengths & weaknesses · Space/Aerospace and Research personas

04

Market Scope & TAM

Market scenario, total available market, and sizing process

05

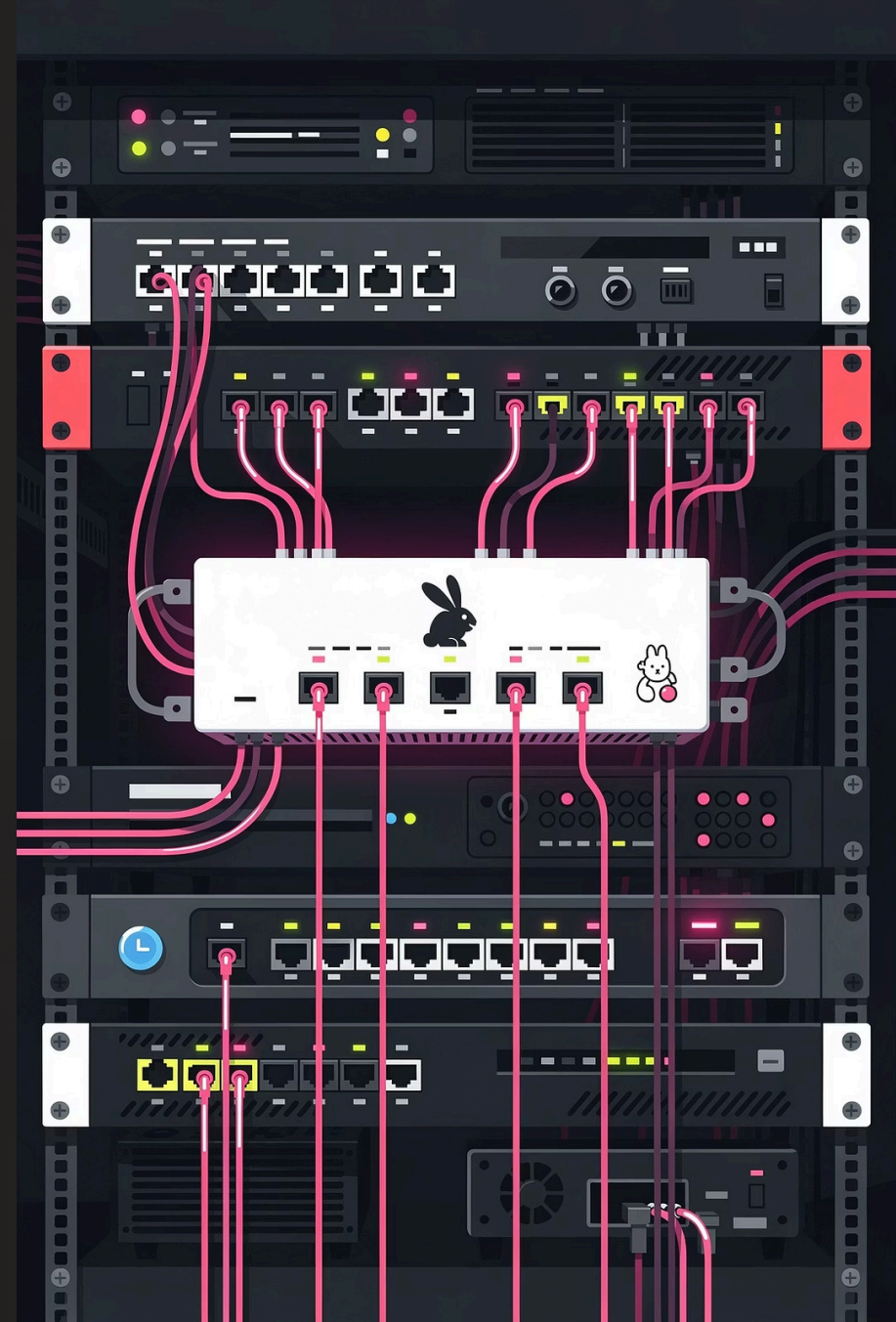
SAM & Way Forward

Serviceable addressable market calculation and strategic implications

White Rabbit Timing System

PRODUCT OVERVIEW

The White Rabbit Timing System delivers sub-nanosecond synchronization over real-time Ethernet networks, developed through a collaborative APNT project with CERN and GSI Helmholtz Centre. The WR-Z16 flagship device offers 16 SFP connectors supporting WR, PTPv2, and NTP protocols, with failover and holdover capabilities optimized for Metro Area Network distances.



White Rabbit: Strengths & Weaknesses

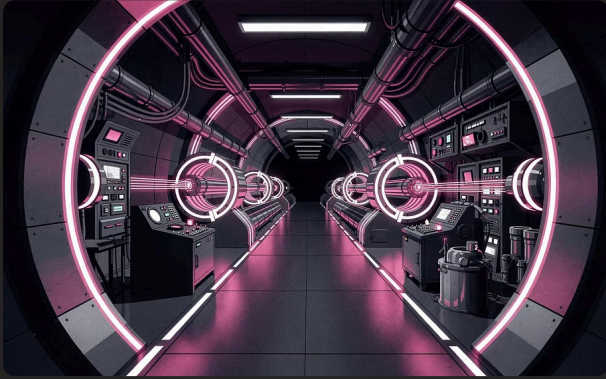
Strengths

- **Collaborative Credibility**
CERN and GSI Helmholtz Centre partnership solidifies reliability and institutional trust.
- **Sub-Nanosecond Precision**
Rivals dedicated hard-wired systems with added Ethernet flexibility and modularity.
- **Standalone WR-Z16**
16 SFP connectors for optical fiber time distribution – no external dependencies.
- **Continuous Innovation**
Updated product lines and strong presence at IPAC and peer industry events.

Weaknesses

- **Premium Pricing**
Higher cost than basic network timing systems may deter budget-constrained clients.
- **Limited Track Record**
As a newer entrant, historical performance data is less extensive than established alternatives.

White Rabbit: Customer Personas



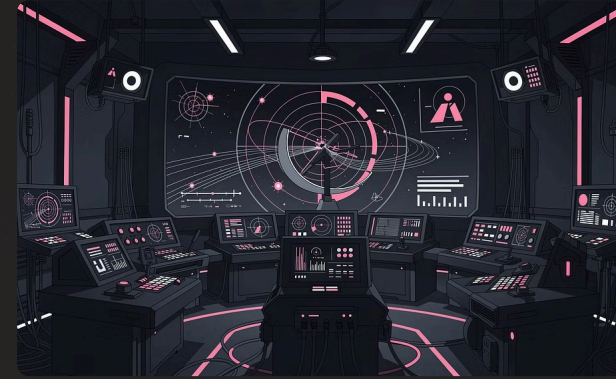
Persona #1 – High Energy Physics

Industry: HEP / Particle Accelerator Facilities

Products: WR-Z16 ·WR ZEN TP-FL ·WR ZEN TP-32BNC

Target Clients: Fermilab, SLAC, UofR LLE, Omega Laser Facility, accelerator OEMs, and research universities

"The WR-Z16 is the top choice – 16 optical timing ports, multi-source time references, Metro Area Network reach, and datacenter-grade failover for precision experiments."



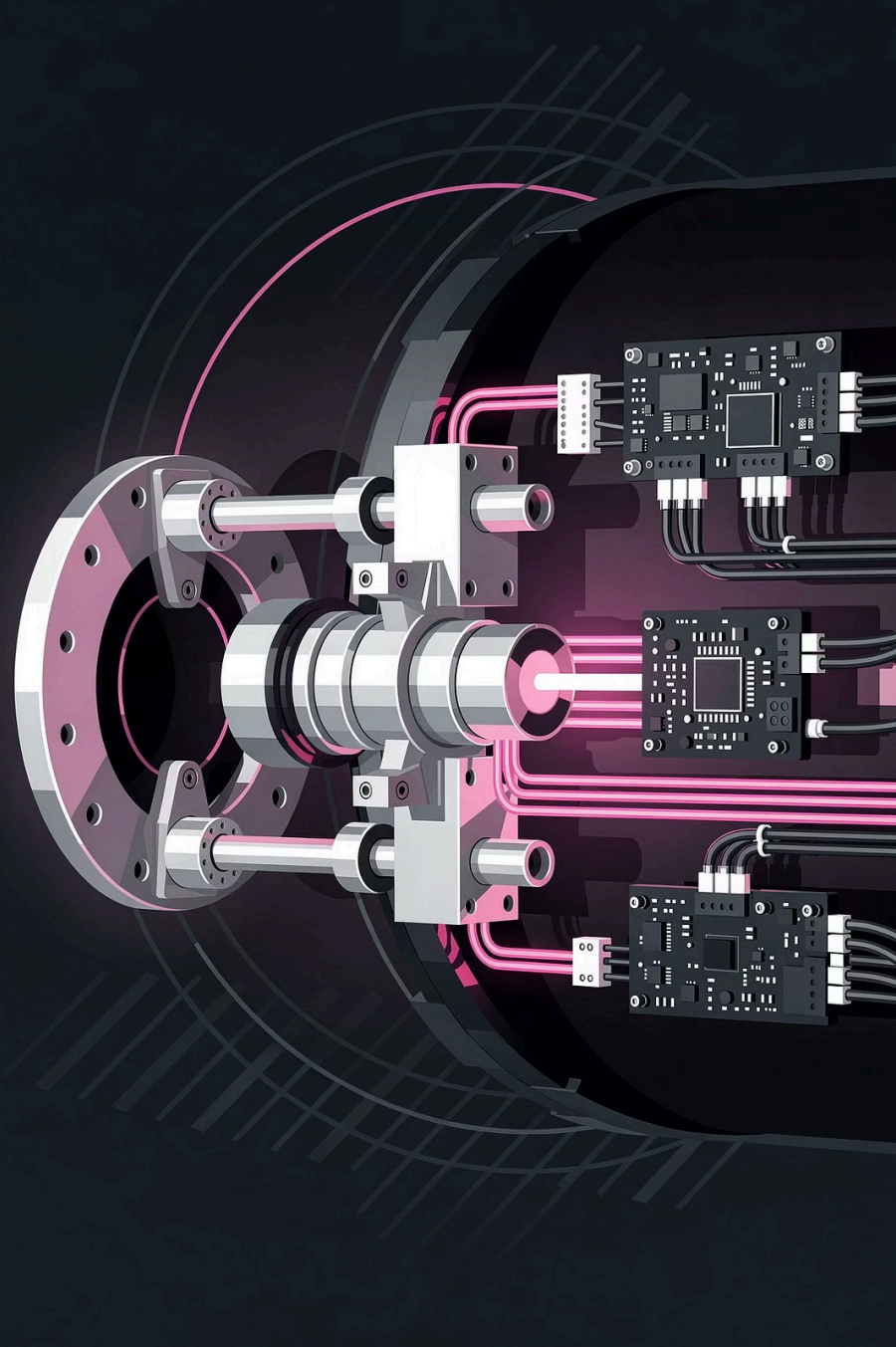
Persona #2 – Military & Aerospace

Industry: Defense / Aerospace Systems Integration

Products: WR-Z16 ·WR ZEN TP-FL ·WR ZEN TP-32BNC

Target Clients: Lockheed Martin, Boeing, Airbus, BAE Systems, Raytheon

"White Rabbit delivers sub-nanosecond precision for radar and network synchronization – essential for operational harmony in the most demanding defense environments."



Beam Position Monitor

PRODUCT OVERVIEW

Safran's Beam Position Monitoring System provides high-accuracy, real-time beam diagnostics leveraging cutting-edge detection technology. Designed for applications ranging from particle accelerator R&D to precision radiation therapy, the system delivers position precision $<250\text{ }\mu\text{m}$, phase precision $<1^\circ$, and alarm response times $<2\text{ }\mu\text{s}$ – critical for both scientific integrity and patient safety.

BPM: Strengths & Weaknesses

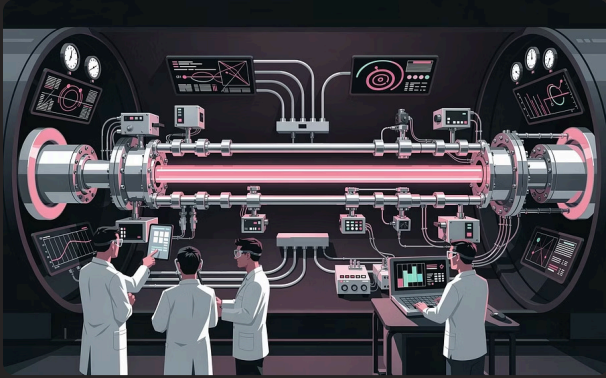
Strengths

- **Advanced Detection Technology**
High accuracy and precision in radiation beam delivery and medical imaging manufacturing.
- **Customizable Architecture**
Tailored configurations for diverse scientific and clinical applications.
- **Real-Time Monitoring**
Sub-2 μ s alarm response enables instant adjustment during active treatment or experiments.
- **Global Support Network**
Safran's worldwide presence ensures reliable post-deployment service and technical assistance.

Weaknesses

- **Higher Cost**
Advanced feature set positions BPM above basic monitoring alternatives on price.
- **Integration Complexity**
Deployment may require additional modifications and specialized installation expertise.
- **Training Requirements**
Effective operation demands specialized knowledge and dedicated staff training programs.
- **Limited Compatibility**
Specific compatibility requirements may restrict applicability in certain legacy setups.

BPM: Customer Personas



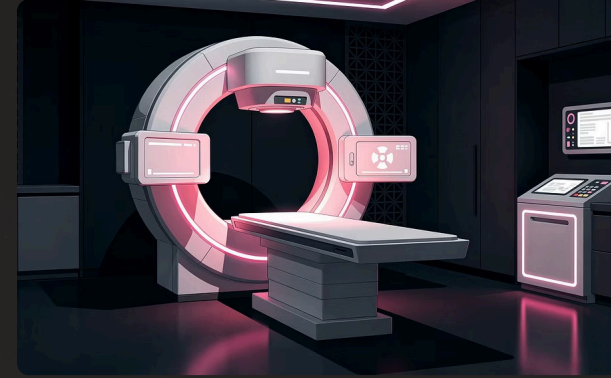
Persona #1 – Scientific R&D Labs

Industry: Research & Development / National Laboratories

Products: UTCA / Stand-Alone BPM

Target Clients: Lawrence Berkeley National Lab, MIT Lincoln Lab, Caltech, Stanford University

"For precision-experiment institutions, our BPM solution provides the accuracy and reliability required to maintain data integrity and ensure scientific success."



Persona #2 – Healthcare / Radiation Therapy

Industry: Oncology / Medical Imaging Device Manufacturing

Products: UTCA / Stand-Alone BPM

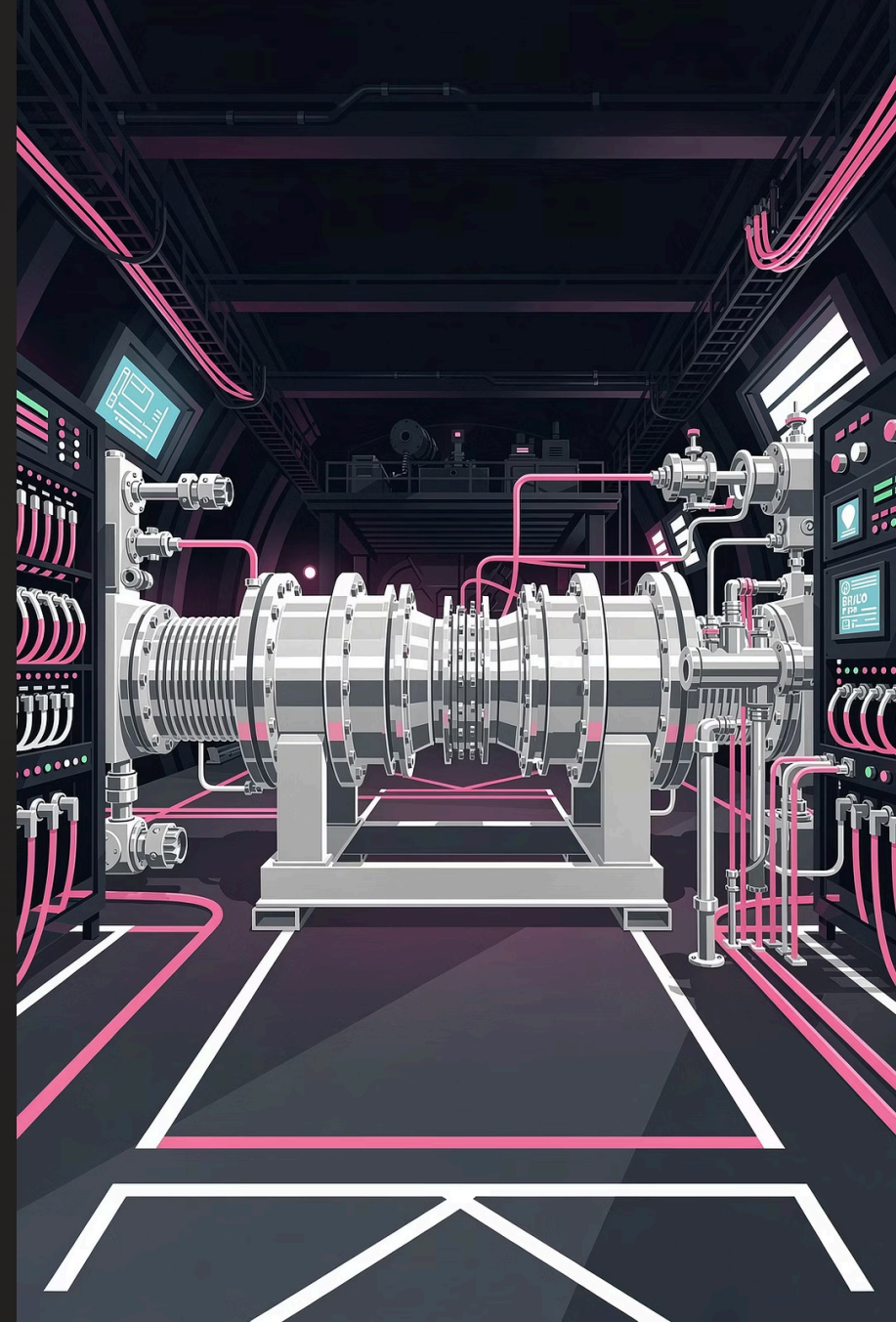
Target Clients: University of Rochester, American Cancer Society, Yale New Haven Hospital, Fox Chase Cancer Center

"Position, phase, and current alarms with $<2 \mu\text{s}$ response, $<250 \mu\text{m}$ position precision, and $<1^\circ$ phase precision – unmatched among competitors in clinical radiation delivery."

Low Level RF (LLRF)

PRODUCT OVERVIEW

Safran's Low Level RF systems provide precise control of accelerating cavities across particle accelerators, space simulators, and advanced manufacturing environments. Backed by deep R&D investment and a globally recognized brand, the UTCA and Stand-Alone LLRF platforms deliver energy and phase control essential for next-generation scientific and space exploration missions.



LLRF: Strengths, Weaknesses & Personas

Strengths

Technological Expertise

Deep LLRF R&D capabilities and a track record of advanced system development.

Integrated Solutions

Comprehensive end-to-end offerings preferred by large institutional clients.

Brand Reputation

Established credibility enhances client confidence in mission-critical deployments.

Global Reach

Worldwide support infrastructure ensures post-deployment reliability.

Personas & Target Clients

Space & Aerospace

Material testing, propulsion R&D, space environment simulation.

Clients: NASA, ESA, Airbus, Boeing

Scientific Research

Particle accelerator control, energy/phase management, materials science.

Clients: Fermilab, CERN, national research institutions

📌 **Key Weakness:** High cost structures and integration complexity may limit adoption in budget-constrained or legacy environments. Specialized market focus can also narrow the accessible customer base.

Market Scenario & Scope

All three product lines address U.S. markets projected to grow substantially through 2027, driven by accelerating demand in scientific research, defense, healthcare, and space exploration. Below are the market parameters for each segment per **Markets and Markets** research.

White Rabbit

\$1.3B → \$1.8B (2022-2027)

CAGR: 6.4%

Driven by telecom, financial services, and energy sector demand for precision timing.

Key competitors: Meinberg, Symmetricom (Microsemi), CAEN, Microchip Technology

BPM

\$1.0B → \$1.5B (2022-2027)

CAGR: 6.5%

Driven by medical and industrial sector demand for beam monitoring accuracy.

Key competitors: Bergoz, Danfysik

LLRF

\$1.0B → \$1.5B (2022-2027)

CAGR: 8.5%

Driven by increasing U.S. government and institutional R&D investment.

Fastest-growing segment among the three product lines.

Total Available Market (TAM)

Across all three product lines and customer personas, Safran has identified a combined universe of over **150 facilities** that represent the total potential addressable market in the United States.

150+

Total Facilities

Identified across all three product lines and customer personas.

70%

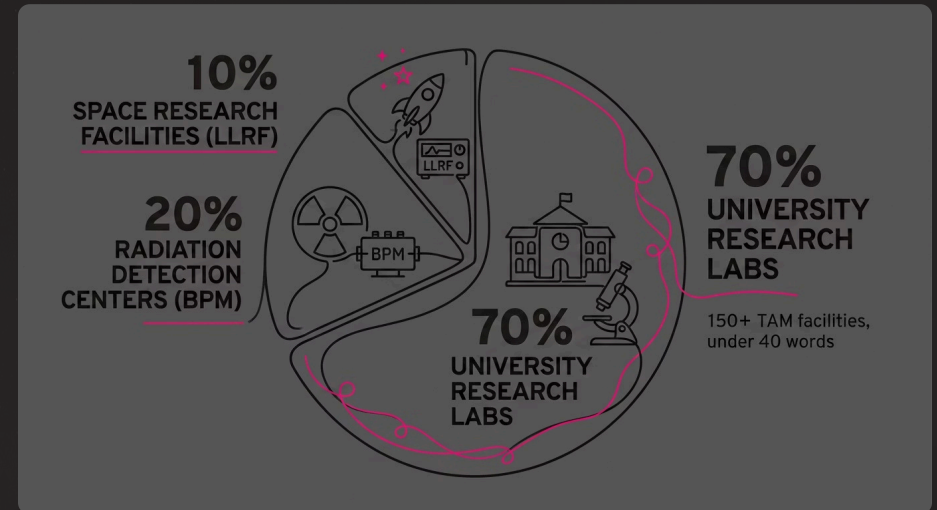
University Labs

Research facilities at universities across the U.S. represent the dominant segment.

30%

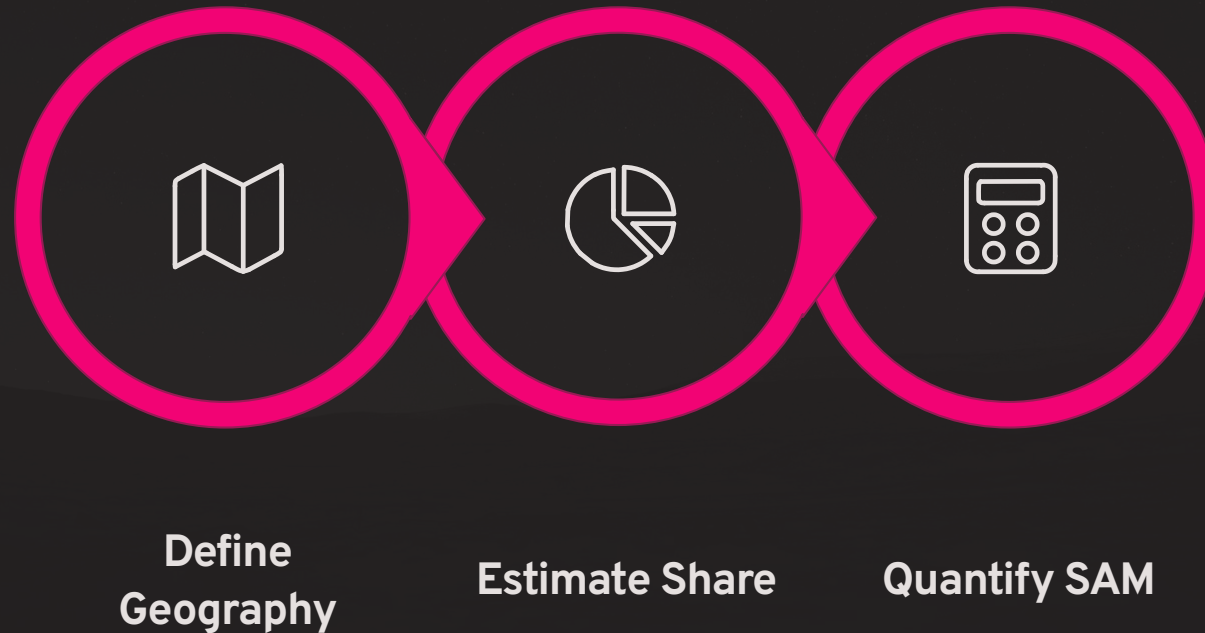
Specialized Centers

Radiation detection centers (BPM) and space research facilities (LLRF).



Process to Size the Market

Translating TAM into actionable targets requires disciplined narrowing through geographic constraints, competitive assessment, and realistic win-rate assumptions.



Geographic segmentation acknowledges regulatory and logistical realities. Competitive share estimates are continuously refined as Safran's value proposition evolves. The SAM formula integrates all three factors into a defensible, strategic market target.

Serviceable Addressable Market (SAM)

METHODOLOGY & CALCULATION

The SAM is derived from a structured formula applied to the identified TAM of 170 facilities (including pipeline prospects), adjusted for demonstrated market interest and competitive win probability.

SAM Formula

$$\text{SAM} = \text{TAM} \times \text{Target Segment\%} \times \text{Win\%}$$

$$\text{SAM} = 170 \times 0.33 \times 0.5 = \mathbf{28 \text{ facilities}}$$

Target Segment %: From 12 sampled facilities, 8 qualified – yielding 66%. Adjusted to 33% given lab-heavy respondent bias.

Win %: Conservative 50% assumed, reflecting competitive landscape and Safran's 80% European success rate as a benchmark.

Definition of Terms

TAM

Total market demand across all 170 identified potential customers.

Target Segment %

Share of sampled facilities showing active product interest (adjusted for sample bias).

Win %

Conservative capture rate accounting for brand strength and competitive pressure.

SAM Key Takeaways

The calculated SAM of **28 facilities** represents the near-term, realistically capturable market share for Safran's three product lines in the U.S. This figure is grounded in empirical sampling data and conservative competitive assumptions.

1

SAM = 28 Facilities

A concrete, actionable target derived from TAM, segment interest, and brand win rate – not a theoretical maximum.

2

33% Segment Interest

A significant portion of the sampled market demonstrates active interest, validating the product-market fit hypothesis.

3

50% Win Rate (Conservative)

Accounts for competition from Meinberg, Bergoz, and others. Safran's 80% European win rate suggests upside potential.

4

Continuous Monitoring Required

SAM assumptions must be revisited quarterly as competitive dynamics and Safran's value proposition evolve.

Way Forward

STRATEGIC PRIORITIES

1

Validate SAM

Expand sample study beyond 30 facilities to refine segment percentages and reduce lab-respondent bias.

2

Prioritize Personas

Focus near-term sales efforts on HEP and Scientific R&D labs — highest density and fastest qualification cycle.

3

Strengthen Brand

Leverage CERN/GSI partnerships and IPAC presence to accelerate credibility with U.S. procurement managers.

4

Track & Adapt

Continuously reassess win rate assumptions against U.S. pipeline performance and European benchmarks.

